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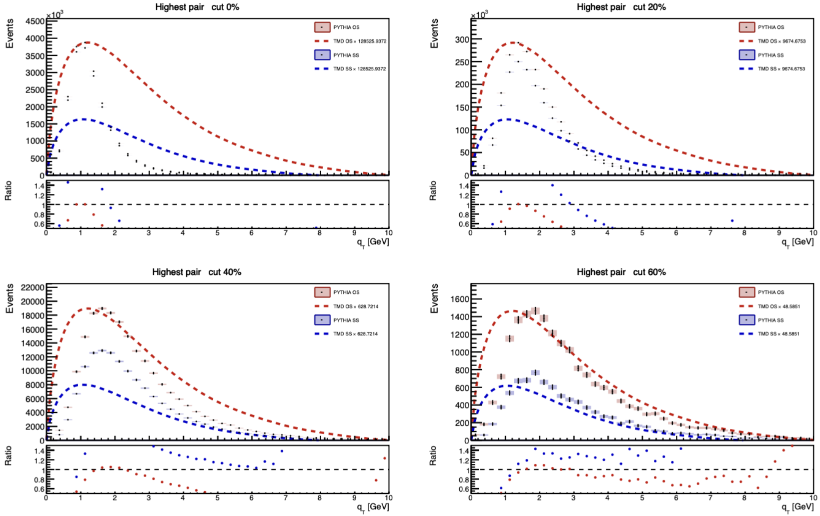
Final project meeting!!!
Week 11

A TMD-oriented analysis of $\pi^+\pi^-$ pairs in e^+e^- collisions

Outline

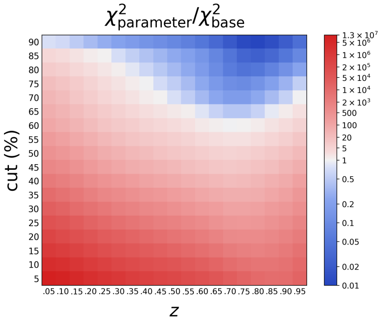
- Recap from last week.
- Best-z scan.
- Leading-order scan.
- PYTHIA parameter scan.
- Best combination.

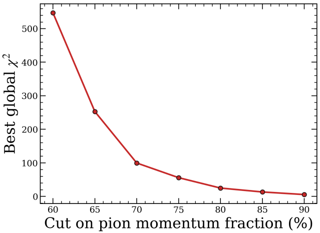
Recap from last week



- Good OS/SS ratio signal, and the theory starts to match reasonably well for $q_T \gtrsim 1.25$ GeV.
- Combined χ^2 for OS and SS is about 270 using all bins, and about 9 after removing the first 5 bins.
- Main goal now is to test 3 optimisation methods, then combine them and see if that improves

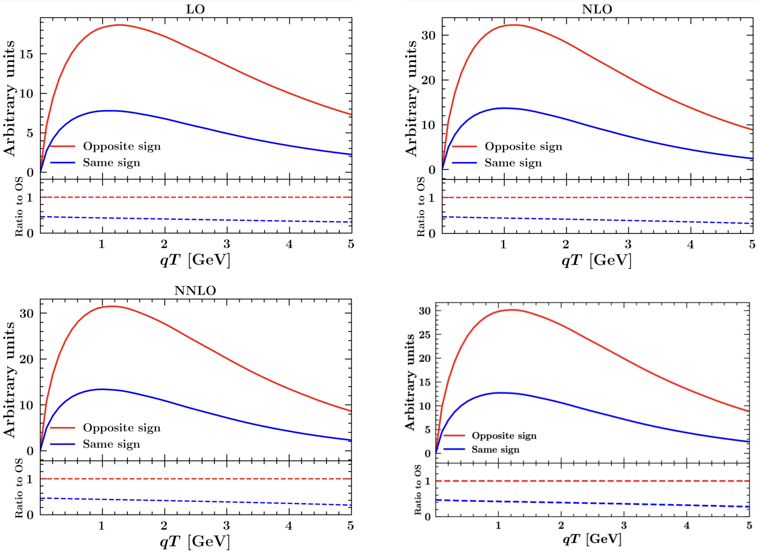
Optimising the comparison in cut and z





- The broadest view is the z versus cut heatmap, which shows the comparison improving into the high-cut, high-z region.
- The best global χ^2 falls strongly as the cut is tightened.
- The very highest cuts can look artificially good because the accepted event count becomes too small.
- The goal here is to identify the most informative region, not to claim a statistically good global fit.

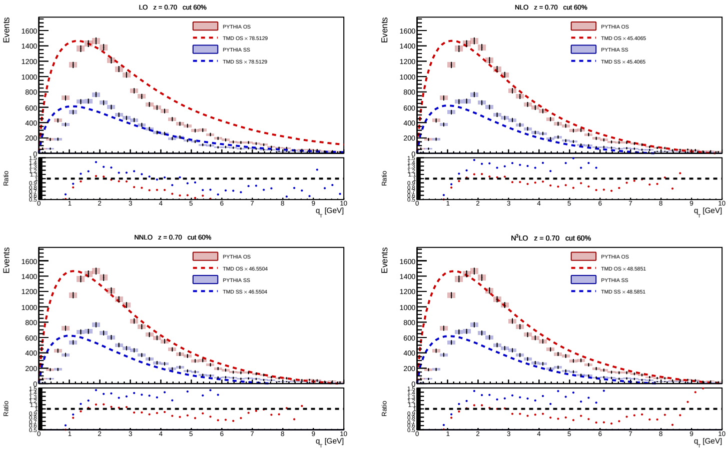
Leading-order scan



- No big difference in shape across the different perturbative orders.
- The comparison does improve slightly in χ^2, but not dramatically.

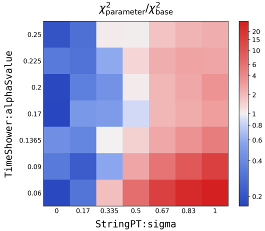
Order	Global χ^2
LO	632.888
NLO	597.877
NNLO	616.954
Current	547.259

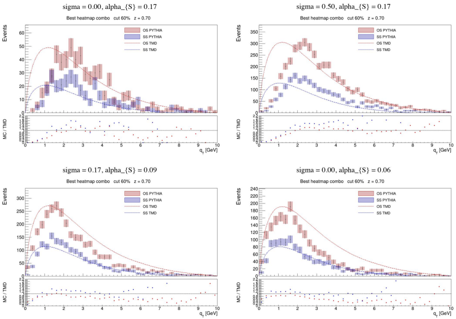
Theory-order comparison



- This compares LO, NLO, NNLO and N3LO theory overlays for the 60% cut at $z = 0.70$.
- No single order gives a fully satisfactory fit.
- The comparison is still useful because the Monte Carlo hard process is only leading order, while the theory can be taken much higher.

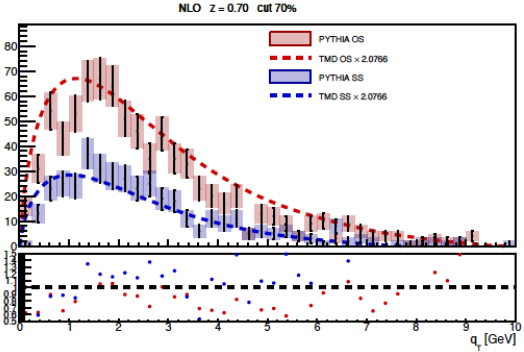
PYTHIA parameter scan and selected points





- White square is the default PYTHIA values, for 20 million counts at $z = 0.70$ and the 60% cut.
- We tested the best-value region, but the error bars became too large.
- Changing the parameters also changes how many successful events are counted.
- So I selected a few points that were blue in the heatmap while still giving decent event counts

Best combination of all 3



- Best combination of all 3 optimisation methods.
- Global χ^2 using all bins:

$$9.79277$$

- This is the best final result.

Thank you!